

SHORT BIO. My research focuses on causal decision making (NeurIPS-25, UAI-26, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (UAI-26), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATION

Graduate School of Data Science, Seoul National University Seoul, Republic of Korea
 Integrated M.S. and Ph.D. in Data Science Sep. 2022 – Feb. 2027
 Advisor: Prof. Sanghack Lee.
 Thesis: “Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals”.
 Overall GPA: 4.02 / 4.3; Ph.D. Qualification Exam partially waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University Seoul, Republic of Korea
 B.S. in Applied statistics and Mathematics (Double major), *Cum Laude* Mar. 2016 – Aug. 2022
 Advisors: Prof. Ji Eun Lee and Jin Hee Yoon. (2.5 years military service)
 Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint authorship and † for corresponding author.

- ◇ Min Woo Park and Sanghack Lee†. [On Transportability for Structural Causal Bandits](#). *Uncertainty in Artificial Intelligence* (UAI), 2026.
- ◇ Min Woo Park*, Taehui Yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, and Sanghack Lee†. Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery. *Uncertainty in Artificial Intelligence* (UAI), 2026.
- ◇ Yesong Choe, Yeahoon Kwon*, Min Woo Park*, and Sanghack Lee†. Canonical Domain Reduction for Partial Counterfactual Identification. *Uncertainty in Artificial Intelligence* (UAI), 2026.
- ◇ Min Woo Park and Sanghack Lee†. [Counterfactual Structural Causal Bandits](#). *International Conference on Learning Representations* (ICLR), 2026.
- ◇ Min Woo Park, Andy Ardit, Elias Bareinboim†, Sanghack Lee†. [Structural Causal Bandits under Markov Equivalence](#). *Neural Information Processing Systems* (NeurIPS), 2025, Columbia CausalAI Laboratory, Technical Report (R-122)
- ◇ Yonghan Jung†, Min Woo Park, and Sanghack Lee†. [Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference](#). *Neural Information Processing Systems* (NeurIPS), 2024.
- ◇ Min Woo Park*, and Ji Eun Lee*†. [Computation of the iterated Aluthge, Duggal, and Mean transforms](#). *Filomat*, 2023.
- ◇ Ji Eun Lee*† and Min Woo Park*. Convergence of the iterated mean transforms of a 2×2 matrix. under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship Spring 2024, Spring 2025, Spring 2026

BK21 scholarship Fall 2024, Fall 2025

First place merit scholarship | Sejong University Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University
 ▷ Mitigating spurious regression in time-series data via Granger causality. Nov. 2021

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University
 ▷ The significance of the Euler’s number and its applications. Nov. 2020

RESEARCH EXPERIENCE

Furiosa AI AI Research Engineer Internship	Jun. 2026 – Nov. 2026
Causality Lab, Seoul National University Ph.D. Researcher	Jan. 2023 – Feb. 2027
Functional Analysis Lab, Sejong University Undergraduate Researcher	Jul. 2021 – Dec. 2022
ISDS Lab, Sejong University Undergraduate Researcher	Jan. 2020 – Dec. 2022

PROJECTS

Towards More Scalable Score-based Causal Discovery LG AI Research	Sep. 2025 – Aug. 2026
▷ Leading a project on scalable score-based causal discovery algorithms. - Developed a novel operator enabling score-based causal discovery algorithms to escape local optima. - Investigating parallelization strategies for score-based causal discovery algorithms. Resulting paper accepted at UAI 2026.	
Hyundai NGV Data Boot-Campus 2025 Hyundai Motor Company	Jul. 2025 – Sep. 2025
▷ Developed high-voltage battery performance prediction technology.	
Hyundai NGV Data Boot-Campus 2024 Hyundai Motor Company	Jul. 2024 – Sep. 2024
▷ Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis.	
Deep Reinforcement Learning for Recommendation Systems Coursework Project	Mar. 2024 – Jun. 2024
▷ Developed a recommendation system based on DDPG, leveraging multi-modal item features (video, audio, and text) to improve recommendation performance.	
Optimal Mixed Policies Completeness Oxford University	Aug. 2023 – Feb. 2024
▷ Studied characterization of conditions under which an intervention effect of a variable is positive. Acknowledged for helpful comments in Toward a Complete Criterion for Value of Information in Insoluble Decision Problems. <i>Transactions on Machine Learning Research</i> (TMLR), 2024.	

TEACHING ASSISTANTS

Causal Inference for Data Science	Spring 2024, Fall 2024, Spring 2025, Spring 2026
▷ Topics included Bayesian networks and graphical causal inference.	
Machine Learning and Deep Learning for Data Science II	Fall 2023
▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.	
Basic Mathematics for Artificial Intelligence (Undergraduate)	Spring 2022
▷ Covered linear algebra, basic statistics, and probability theory.	

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Academic Services

Conference Reviewer: **2026** NeurIPS, UAI, ICLR, ICML, **2025** ICML
Reviewed AI/ML lecture video content for LG AI Research AIMERS
Summer 2025, Winter 2025, Summer 2026

Invited Talks

SNU GSDS student research seminar | Seoul, Republic of Korea
Dec. 2024, Apr. 2025

Poster Sessions

UAI 2026 poster session | Amsterdam, Netherlands
Aug. 2026 (scheduled)
ICLR 2026 poster session | Rio de Janeiro, Brazil
Apr. 2026
ExploreCSR poster session | Seoul, Republic of Korea
Dec. 2025
NeurIPS 2025 poster session | San Diego, USA
Dec. 2025
JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea
Nov. 2025
SNU GSDS workshop poster session | Jeju, Republic of Korea
Jul. 2025
NeurIPS 2024 poster session | Vancouver, Canada
Dec. 2024

REFERENCE

Sanghack Lee | Graduate School of Data Science, Seoul National University
Associate professor

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Ji Eun Lee | Department of Mathematics and Statistics, Sejong University
Associate professor

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Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University
Associate professor

jin9135@sejong.ac.kr

MENTORING

Taehui Yun | Graduate School of Data Science, Seoul National University
Ph.D. student. Co-advised a project; paper accepted at UAI 2026.

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